

A Reproducible Kerr Force-Free Split-Monopole Gate

A Negative Finite-Spin Search Result and a Positive Blandford-Znajek Perturbative Anchor

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Lagra Project / pde-engine adapter note

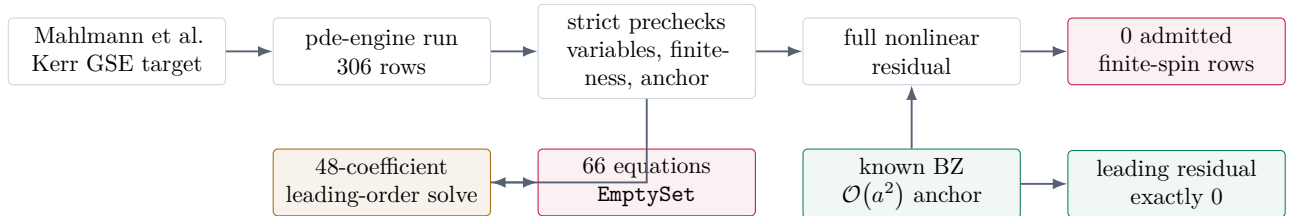
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Abstract

We report a reproducible admission gate for candidate analytic solutions of a closed Kerr force-free split-monopole Grad-Shafranov target. The gate uses a bounded pde-engine expression run, strict symbolic prechecks, a direct nonlinear residual check, targeted finite-spin correction grammars, and a leading-order coefficient solve. The finite-spin search result is negative: 878 candidate inputs were scanned, including 456 targeted finite-spin correction candidates and 112 Kerr-metric-resummed correction candidates that passed the strict prechecks before the full residual, and zero candidates had exact-zero nonlinear residual. A 48-coefficient leading-order ansatz produced 66 polynomial equations and SymPy returned `EmptySet`. Separately, the same gate recognizes the known slow-rotation Blandford-Znajek perturbative split-monopole anchor: after substituting the Tanabe-Nagataki/Pan-Yu radial correction and using $\text{Li}_1(z) = -\log(1-z)$, the coefficient of a^2 in the full residual simplifies exactly to zero. This is a calibration of the residual gate, not a claim of a new exact finite-spin Kerr solution.

1 Claim

The strongest result here is negative. Under the implemented closed split-monopole Kerr force-free target, the current bounded pde-engine search and the targeted finite-spin correction families do not produce an admitted exact finite-spin candidate. The positive result is narrower but useful: the residual implementation is not vacuous, because it accepts the established $\mathcal{O}(a^2)$ Blandford-Znajek perturbative split-monopole correction as a leading-order anchor.



2 Literature Target

The target equation is the stationary, axisymmetric, relativistic Grad-Shafranov equation for force-free Kerr magnetospheres. Mahlmann, Cerda-Duran, and Aloy describe numerical solution techniques for this equation in Kerr spacetimes and use established configurations such as split-monopole, paraboloidal, black-hole-disk, and uniform field setups [1]. The reason this is a legitimate paper target is not that the equation is new; it is that exact analytic finite-spin Kerr force-free solutions remain

scarce. Camilloni, Grignani, Harmark, Oliveri, and Orselli state the stronger boundary used here: in the extreme Kerr case, there is no known exact analytic solution to force-free electrodynamics that is stationary, axisymmetric, and magnetically dominated [2].

The gate closes the target with the split-monopole potential functions

$$r_+ = M + \sqrt{M^2 - a^2}, \quad (1)$$

$$\Omega(\Psi) = \frac{a}{2(r_+^2 + a^2)}, \quad (2)$$

$$I(\Psi) = -\frac{\Omega}{2}\Psi(2 - \Psi), \quad (3)$$

$$II'(\Psi) = I(\Psi)\frac{dI}{d\Psi}. \quad (4)$$

We write $x = \cos \theta$ and denote the implemented closed residual by

$$\mathcal{R}[\Psi] \equiv \text{GSLightCylinder}[\Psi; \Omega(\Psi), I(\Psi)]. \quad (5)$$

The exact expanded symbolic implementation is in `tools/export_kerr_paper_target_gate.py`, in the function `full_kerr_split_monopole_residual`.

3 Admission Gate

A row is admitted only if it passes all implemented gates below. The order matters. Cheap symbolic filters run before the full nonlinear residual, so the expensive residual simplification is not used to compensate for malformed candidate expressions.

1. The expression depends on r , x , and a .
2. Denominators are finite on the rational safe points.
3. The expression itself is finite on the same points.
4. The $a \rightarrow 0$ limit is the Schwarzschild split-monopole anchor $1 - x$ or the equivalent orientation x .
5. The expression is not exactly one of the known anchors.
6. The closed nonlinear residual $\mathcal{R}[\Psi]$ is exactly zero.

The implemented gate deliberately does not yet prove global horizon, axis, and light-surface regularity. It also rejects exact known anchors but does not yet implement a broad equivalence theory over gauge choices, reparameterizations, and coordinate changes. Those omissions matter for any future positive claim; they do not weaken the negative statement made here.

4 Bounded Search and Correction Families

The source run was

run id	paper_repro_20260526_104304_2b096bdc
problem	kerr_magnetosphere
max depth	2
validators	1
per-expression timeout	3.0 seconds
engine rows loaded	306
completed validations	306

The exact reproduction command recorded by the artifact is

```
python3 tools/export_kerr_paper_target_gate.py \
--run-engine --max-depth 2 --validators 1 \
--timeout-s 90 --validation-timeout-s 3 --max-rows 1000
```

The first targeted finite-spin grammar was

$$\Psi = 1 - x + a^2 c b(r, x), \quad c \in \left\{ -1, -\frac{1}{2}, \frac{1}{2}, 1 \right\}, \quad (6)$$

where $b(r, x)$ is a Cartesian product of eight angular factors and six radial factors. This produced 192 one-term candidates. The two-term screen was

$$\Psi = 1 - x + a^2 (c_1 b_i(r, x) + c_2 b_j(r, x)), \quad c_1, c_2 \in \{-1, 1\}, \quad (7)$$

using a smaller 12-function sub-basis and unordered pairs. This produced 264 two-term candidates. All 456 targeted finite-spin corrections passed the strict prechecks before the full residual. None passed the full residual.

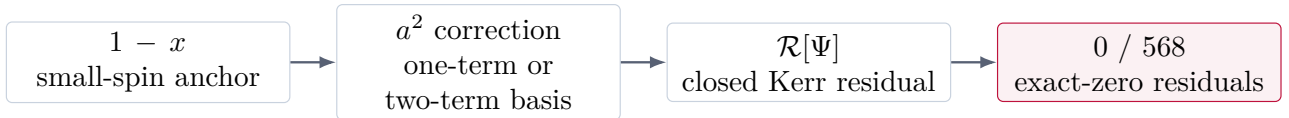
The follow-up search added a Kerr-metric-resummed screen. It preserves the same $a \rightarrow 0$ split-monopole anchor but permits finite-spin denominators built from the metric factors

$$\Sigma = r^2 + a^2 x^2, \quad \Delta = r^2 - 2Mr + a^2, \quad A = (r^2 + a^2)^2 - \Delta a^2 (1 - x^2). \quad (8)$$

The screen has the form

$$\Psi = 1 - x + a^2 c m(r, x, a, M), \quad c \in \left\{ -1, -\frac{1}{2}, \frac{1}{2}, 1 \right\}, \quad (9)$$

with four anchor-preserving angular factors and seven metric radial/denominator factors. This generated 112 additional candidates. All 112 passed the strict prechecks before the full residual. None passed the full residual.



5 Coefficient Solve

The sampled coefficients above are only a finite screen. To avoid mistaking that for a proof about the one-term basis, the gate also solves the leading coefficient system

$$\Psi = 1 - x + a^2 \sum_{i=1}^{48} c_i b_i(r, x), \quad M = 1, \quad (10)$$

and expands the residual as

$$\mathcal{R}[\Psi] = a^2 P(r, x; \mathbf{c}) + \mathcal{O}(a^4). \quad (11)$$

After denominator clearing, the polynomial identity $P = 0$ gives 66 equations in 48 unknowns. SymPy reports

$$\text{linsolve}(A\mathbf{c} = 0) = \emptyset. \quad (12)$$

Thus the one-term ansatz family has no leading-order correction within this basis; it is not merely that four sampled coefficient values failed.

6 Perturbative Blandford-Znajek Anchor

The calibration screen inserts the known slow-rotation split-monopole correction from the Blandford-Znajek perturbative literature. Tanabe and Nagataki analyze higher-order terms for the Kerr-parameter expansion and emphasize that the known analytic monopole solution is perturbative through second order in spin [3]. Pan and Yu extend the split-monopole perturbation solution to fourth order and likewise identify the second-order split-monopole solution as the known analytic Blandford-Znajek baseline [4].

With $M = 1$, the gate inserts

$$\Psi_{\text{BZ}} = 1 - x + a^2 x(1 - x^2)R(r), \quad (13)$$

where

$$\begin{aligned} R(r) = & \frac{r^2(2r-3)}{8} \left[\text{Li}_2\left(\frac{2}{r}\right) - \log\left(1 - \frac{2}{r}\right) \log\left(\frac{r}{2}\right) \right] \\ & + \frac{1+3r-6r^2}{12} \log\left(\frac{r}{2}\right) + \frac{11}{72} + \frac{1}{3r} + \frac{r}{2} - \frac{r^2}{2}. \end{aligned} \quad (14)$$

The check is

$$[a^2] \mathcal{R}[\Psi_{\text{BZ}}] = 0, \quad (15)$$

where the residual is series-expanded through $\mathcal{O}(a^4)$. The symbolic normalization uses the identity

$$\text{Li}_1(z) = -\log(1-z). \quad (16)$$

After that replacement, the leading residual simplifies exactly to zero, and all rational point checks are zero. The artifact records this as `passes_leading_order_anchor`. It also records `finite_spin_exact_candidate=false`. That flag is not cosmetic: Ψ_{BZ} is a perturbative $\mathcal{O}(a^2)$ anchor and must not be reported as a new exact finite-spin solution.

7 Results

The machine-readable artifact status is `no_candidate_yet`. That status is not a failure of the gate; it is the correct output when the gate is implemented and no candidate in scope passes it.

Screen	Status	Evidence
engine generation	implemented	306 generated rows loaded for the strict gate.
strict prechecks	implemented	candidate variables, finite denominators, finite values, small-spin anchor, and exact-anchor rejection recorded per row.
full nonlinear residual	implemented negative	0 exact-zero residual rows; admitted count 0.
bounded correction screens	implemented negative	192 one-term and 264 two-term corrections; 456 passed strict prechecks; 0 passed the full residual.
metric-resummed screen	implemented negative	112 Kerr-metric-denominator corrections; 112 passed strict prechecks; 0 passed the full residual.
coefficient solve	no leading-order solution	66 equations, 48 unknowns, matrix shape [66,48], <code>EmptySet</code> .
literature perturbative anchor	passes leading-order anchor	Tanabe-Nagataki/Pan-Yu $\mathcal{O}(a^2)$ correction gives exact-zero leading residual.
equivalence filters	partial	exact known anchors rejected; broad gauge/reparameterization equivalence remains future work.
global regularity	not sufficient for positive claim	rational safe-point checks exist; global horizon, axis, and light-surface proofs are not implemented.

The paper result is therefore:

finite-spin exact Kerr candidate: not found

known $\mathcal{O}(a^2)$ BZ perturbative anchor: recovered

(17)

8 Reproducibility

The machine-readable source of truth is `docs/kerr-paper-target-gate.json`. The human-readable gate summary is `docs/kerr-paper-target-gate.md`. The implementation and tests are:

- `tools/export_kerr_paper_target_gate.py`
- `tests/test_kerr_paper_target_gate.py`

The relevant regression asserts that the literature slow-rotation anchor is enabled, has status `passes_leading_order_anchor`, has `leading_residual_exact_zero=true`, has simplified residual 0, and is not a finite-spin exact candidate.

The verification bundle used for the current artifact included Python syntax checks, the force-free and Kerr unit tests, JSON validation, PDF text checks, the Lean normalizer build, and Git whitespace hygiene. The generated SQLite run outputs and validator caches are intentionally not part of the paper claim; they are local execution artifacts.

9 Conclusion

The work converts an ambiguous search program into an executable scientific gate. The gate currently rejects every generated or targeted finite-spin candidate in scope. This is a meaningful negative result because the same residual implementation accepts the known Blandford-Znajek slow-rotation perturbative correction at leading order. The next publishable positive paper would need a candidate that passes the exact residual, stronger equivalence filters, and global regularity checks. This paper does not claim that such a candidate has been found.

References

- [1] J. F. Mahlmann, P. Cerda-Duran, and M. A. Aloy, *Numerically solving the relativistic Grad-Shafranov equation in Kerr spacetimes: numerical techniques*, Monthly Notices of the Royal Astronomical Society 477, 3927–3945 (2018), [arXiv:1802.00815](#).
- [2] F. Camilloni, G. Grignani, T. Harmark, R. Oliveri, and M. Orselli, *Moving away from the Near-Horizon Attractor of the Extreme Kerr Force-Free Magnetosphere*, JCAP 10, 048 (2020), [arXiv:2007.15665](#).
- [3] K. Tanabe and S. Nagataki, *Extended monopole solution of the Blandford-Znajek mechanism: Higher order terms for a Kerr parameter*, Physical Review D 78, 024004 (2008), [doi:10.1103/PhysRevD.78.024004](#), [arXiv:0802.0908](#).
- [4] Z. Pan and C. Yu, *Fourth-order split monopole perturbation solutions to the Blandford-Znajek mechanism*, Physical Review D 91, 064067 (2015), [doi:10.1103/PhysRevD.91.064067](#), [arXiv:1503.05248](#).